Physics / Secondary

Force, mass and acceleration

A resultant force changes an object's velocity. In a simple constant-mass model, resultant force equals mass multiplied by acceleration.

Core ideas

1. Use F = ma with force in newtons (N), mass in kilograms (kg) and acceleration in m/s².

2. The resultant force is the vector sum of the forces acting on the object.

3. Zero resultant force means zero acceleration. An object may still be moving at a constant velocity.

Worked example

A 3 kg object has a resultant force of 12 N. Acceleration = F / m = 12 / 3 = 4 m/s².

Try it yourself

1. Find the force needed to accelerate 5 kg at 2 m/s².

2. Find the acceleration of 4 kg under a resultant force of 6 N.

3. Does zero resultant force always mean an object is stationary?

Check your answers

1. 10 N

2. 1.5 m/s²

3. No. It may move with constant velocity.